

Determinants of Emotion Duration and Underlying Psychological and Neural Mechanisms

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Abstract

Emotions are traditionally considered to be brief states that last for seconds or a few minutes at most. However, due to pioneering theoretical work of Frijda and recent empirical studies, it has become clear that the duration of emotions is actually highly variable with durations ranging from a few seconds to several hours, or even longer. We review research on determinants of emotion duration. Three classes of determinants are identified: features related to the (a) emotion-eliciting event (event duration and event appraisal), (b) emotion itself (nature of the emotion component, nature of the emotion, and emotion intensity), and (c) emotion-experiencing person (dispositions and emotion regulatory actions). Initial evidence on the psychological and neural mechanisms that underlie their effects is discussed.

Keywords

affective chronometry, affective neuroscience, determinants, emotion duration, emotion dynamics

Emotions are processes that unfold over time. Consequently, a full theoretical understanding of how they operate can only be reached when their time-related aspects are captured adequately (Davidson, 1998; Frijda, 2007). Moreover, research on emotion dynamics is important from a clinical perspective as disturbances in emotion dynamics are symptomatic of several mental disorders such as depression (Siegle, Steinhauer, Thase, Stenger, & Carter, 2002) and may lead to the development of somatic diseases such as cardiovascular illness (Brosschot, Gerin, & Thayer, 2006).

As such, it is surprising that for a long time the dynamic properties of emotions were largely ignored. This partially results from the traditional belief that emotions are activity bursts of brief duration, lasting for seconds or a couple of minutes at most (i.e., emotions as brief fight/flight responses). Consistent with this belief, researchers identified the nature and (initial) intensity of such outbursts and no further questions regarding emotion dynamics were posed (Frijda, 2007). In his opponent-process theory of motivation, Solomon already challenged this approach four decades ago and noticed that the

Author note: This research was supported in part by the Research Fund of KU Leuven (GOA/15/003). Philippe Verduyn is a Postdoctoral Fellow of the Research Foundation – Flanders (FWO).

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duration of emotions is actually highly variable (Solomon & Corbit, 1974). However, systematic empirical evidence to support this claim was lacking, and emotions remained perceived as states of brief duration by the scientific community (Ekman, 1984). Another 20 years passed before Frijda and colleagues (Frijda, Mesquita, Sonnemans, & van Goozen, 1991) argued and empirically showed that emotions are generally not flash-like responses but typically last for several minutes, hours, or even longer. Around the same time, other researchers came to the same conclusion and provided additional empirical support (e.g., Gilboa & Revelle, 1994; Scherer & Wallbott, 1994). More recently, a series of diary studies confirmed that an emotion can last anywhere from a couple of seconds up to several hours (Verduyn, Delvaux, van Coillie, Tuerlinckx, & van Mechelen, 2009; Verduyn, van Mechelen, Kross, Chezzi, & van Bever, 2012; Verduyn, van Mechelen, & Tuerlinckx, 2011).

The realization that emotions are complex dynamic processes and that their duration shows a remarkable broad range poses emotion researchers for the challenge to account for this variability. Davidson (1998) created a research field that aims to get a better understanding of variability in emotion dynamics including the duration of emotional recovery, and named it *affective chronometry*. The aim of this article is to provide a review of research on affective chronometry, centering on the duration of emotions. In particular, building on a preliminary review (van Mechelen, Verduyn, & Brans, 2013) research on determinants of emotion duration will be discussed. For each determinant we will theorize on its influence on emotion duration and discuss empirical findings including initial evidence on the psychological and neural mechanisms that underlie their effects. For some determinant classes the number of neurophysiological findings is limited as the time dimension of the emotional response is often ignored in neuroimaging research, particularly when using fMRI (Waugh & Schirillo, 2012). However, within the field of affective neuroscience, consensus is growing that time should take a central place in the study of emotions (e.g., Schuyler et al., 2012; Waugh, Shing, & Avery, 2015).

We start this review by defining emotion duration as the amount of time that elapses between the beginning and end point of an emotional episode. The beginning of an emotional episode can rather easily be identified as, in contrast to moods,¹ emotions start with the occurrence of an external or internal event (even though the start of the emotion does not necessarily exactly coincide with the start of the event; Frijda, 2007). The end of an emotional episode corresponds with the moment that the intensity of the emotional response returns to zero or to a baseline level, either for the first time (Verduyn et al., 2011), for several consecutive times (Sbarra, 2006), or permanently (Frijda, 2007). Even though duration definitions and associated measurements differ across studies, the first definition (i.e., first return to zero or baseline) is most often used.

Determinants of Emotion Duration

Determinants of emotion duration can be grouped into three categories: (a) characteristics involving the event (stimulus) that

elicited the emotion, (b) characteristics of the emotion itself, and (c) characteristics involving the subject who experiences the emotion.

Characteristics of the Emotion-Eliciting Event

Two determinants fall within this category: the duration of the emotion-eliciting event, and perceived event features as captured by appraisal dimensions. Despite that appraisals constitute an element of an emotional response, they are added to this determinant category as the emotion-eliciting event plays a central role in the appraisal process, and because appraisals are considered to cause emotion in addition to being part of it (Moors, Ellsworth, Scherer, & Frijda, 2013).

Event duration. Frijda (2007) conjectured that the duration of the eliciting event is positively related to the duration of the emotional response. This claim was confirmed by Verduyn and colleagues who further showed that reappearances of the elicitor may lead to emotion reactivations, thereby hampering the recovery process (Verduyn et al., 2009; Verduyn, van Mechelen, & Frederix, 2012). Such reappearances likely reactivate the original appraisal configuration which in turn prolongs the emotion.

At this point, three remarks have to be made. First, event duration is not solely determined by ongoing physical presence but also by stability in psychological content (e.g., the enduring presence of someone who made an insulting remark does not imply event prolongation if that person rephrases their remark in an inoffensive manner). Second, the positive relation between the duration of the emotion-eliciting event and emotion duration does not imply that emotions always or even typically end at event offset; it rather means that the emotion is unlikely to end as long as the elicitor remains present. Third, one may conjecture that the occurrence of events that are not related to the emotion may influence emotion duration as well (Hollenstein, 2015), for instance by providing emotional counterweight (e.g., receiving good news when feeling sad) or by offering an opportunity for distraction, but systematic research on this is lacking.

Event appraisals. The appraisal dimension that has most consistently been related to emotion duration is the importance of the eliciting event (i.e., motivational or concern relevance): events perceived as highly important lead to longer emotions (Verduyn et al., 2009; Verduyn et al., 2011). Verduyn, van Mechelen, Tuerlinckx, and Scherer (2013) further showed that negative emotions last especially long when the eliciting event and its consequences are perceived to be incongruent with the individual's goals (appraisal of goal-congruence), norms and values (appraisals of unfairness and immorality), and self-ideal (appraisal of self-ideal congruence). As such, a mismatch between an event and one's desires causes negative emotions (Carver, 2015) and the size of this mismatch predicts the duration of the emotional response.

Two psychological mechanisms have been explored to account for the effect of appraisals on emotion duration. A first

mechanism is that appraisals influence emotion intensity with emotions of high intensity requiring a long recovery period. A second mechanism is that certain appraisal configurations initiate the use of particular regulation strategies that, in turn, affect emotion duration. Appraisal–duration relations became somewhat less outspoken when statistically controlling for emotion intensity (Verduyn et al., 2013) or regulation strategies (Brans & Verduyn, 2014) but they remained highly significant, which implies that additional mechanisms underlie the relationship between appraisals and emotion duration.

The neural bases of several of the appraisal dimensions that were found to be related to emotion duration have been identified (for a review, see Brosch & Sander, 2013). The amygdala acts as a concern relevance detector and represents the neural mechanism through which event importance is established. Following relevance detection, the amygdala orchestrates the further processing of the encountered event. This includes a refinement of the initial appraisal in terms of additional appraisal dimensions, including goal congruency (anterior cingulate cortex and dorsolateral prefrontal cortex), norm compatibility (superior anterior temporal lobe), and value compatibility (medial prefrontal cortex and dorsal striatum) amongst others (Brosch & Sander, 2013). However, neuroscientific research on the effect of these appraisal dimensions on the temporal unfolding of the emotion beyond the period of emotion elicitation is lacking. Consequently, the neural mechanisms that mediate the relation between initial event appraisal and emotion duration are unknown.

Characteristics of the Emotion

Three determinants of emotion duration fall within this category with the first determinant being situated at the level of the elements that constitute an emotion (i.e., nature of the emotion component), and the others reflecting features of the emotion as a whole (i.e., nature and intensity of the emotion).

Nature of the emotion component. In contemporary emotion theories, emotions are defined to consist of several components, including (a) cognitive components, (b) motivational components or action tendencies, (c) (neuro)physiological components, (d) motor components or expressive behaviors, and (e) feeling components or subjective experiences (Scherer, 2005). Frijda argued that each of these response systems may have a specific duration (Frijda et al., 1991). Several theoretical reasons may account for such duration differences. For example, whereas continued expression in the absence of an observer may become meaningless and sustained elevated physiological activity may become damaging due to allostatic load (McEwen, 2000), meaning making typically takes time and requires sustained activity in the cognitive and associated feeling component of the emotional response (Schön, 2010).

Studies that directly compare the duration of different emotion components are largely lacking, but across studies evidence seems available that particular components remain in general active for a longer time than others. In particular, whereas the

duration of expressive (e.g., facial expression; Ekman & Friesen, 1982), and (neuro)physiological components (e.g., cardiovascular recovery; Fredrickson & Levenson, 1998) seems to rarely exceed a few minutes, feelings and associated cognitions may persist for hours or longer (e.g., Nolen-Hoeksema & Morrow, 1993; Verduyn et al., 2011).

However, these duration differences may partly be due to dissimilarities in research methodology: Emotion expression and (neuro)physiological responses are typically measured in the lab following a weak emotion induction without any long-term consequences (e.g., using affective pictures), whereas the duration of the feeling component is typically measured in daily life following a wide range of emotional situations including events of high personal importance. When, instead of in the lab, the duration of expressive behavior (Bylsma, Croon, Vingerhoets, & Rottenberg, 2011) or cardiovascular recovery (Pieper & Brosschot, 2005) was examined in daily life, durations generally exceeded several minutes. Similarly, when, instead of in daily life, the duration of emotional experience was examined in the lab, durations were not found to exceed a few minutes (Garrett & Maddock, 2001). Future research is needed to get a better understanding of the size of duration differences between emotion components.

Nature of the emotion. Frijda (2007) conjectures that different emotions have different built-in time courses. Scherer and Wallbott (1994) provided data supporting this claim as they found evidence for the following rank order of emotion duration: fear = disgust = shame $\leq$ anger < guilt < joy < sadness. More recently, Verduyn and colleagues reported evidence for (parts of) the same rank order (e.g., Verduyn et al., 2009).

According to Frijda (2007), part of this variability may be accounted for by features of the emotion-eliciting event: Whereas fear and disgust are often elicited by transient events without long-term consequences, the consequences of events causing joy or sadness usually extend in time. Along similar lines, Brans and Verduyn (2014) noticed that the long duration of sadness may be due to the relatively high levels of importance attached to the kinds of events that give rise to this emotion.

The neural mechanisms that account for these duration differences remain unknown. This stands in sharp contrast with the plethora of studies aimed at identifying discrete neural correlates for basic emotions, which has led to mixed findings with some recent meta-analyses arguing in favor (e.g., Vytal & Hamann, 2010) and others against the existence of such neural correlates (e.g., Lindquist, Wager, Kober, Bliss-Moreau, & Barrett, 2012). However, the time dimension is typically missing in these studies which is problematic as the difference in neural signature between emotions may not be a matter of *which* neural regions are involved but *when, and for how long* neural regions become and remain active. A demonstration of this was recently provided in an electroencephalography (EEG) study by Costa et al. (2014) who found differences in the onset speed and duration of neural activity for fear, disgust, happiness, and sadness during the period that immediately followed the emotional stimulus. Interestingly, sadness showed a slow neural onset but

the activity endured for the longest time which is consistent with the observation that sadness is a persistent emotion. Similar studies across larger timescales would add much to our understanding of duration differences between emotions.

Intensity of the emotion. Both overall time-independent measures of intensity (Sonnemans & Frijda, 1994) and measures of the initial intensity of an emotional response (Verduyn et al., 2009) have been found to be related to emotion duration, even when controlling for the importance of the emotion-eliciting event. In a recent study by Waugh, Hamilton, and Gotlib (2010), the neural regions that underlie the intensity–duration relationship were identified. In particular, greater negative emotional intensity was associated with longer duration of activation in regions along the cortical midline associated with self-referent processing: the anterior medial prefrontal cortex, and the posterior cingulate cortex (Fossati et al., 2003). As such, following high intense emotional events, people seemingly tend to elaborate on the self-relevance of the event which may cause emotions to endure.

Characteristics of the Emotion-Experiencing Person

Dispositions. Individuals differ in their ability to recover from negative events or to maintain positive feelings (Davidson, 1998). Dispositions have been identified that account for this variability including global trait dimensions within the healthy domain as well as measures of psychopathology. Next, we describe dispositions that have repeatedly been shown to be related to emotion duration.

Resilience. Resilience refers to the ability to cope effectively and adapt in the face of loss, hardship, or adversity (Block & Kremen, 1996). As such, resilience is by definition closely related to the concept of emotion duration. Empirical studies provide support for this with resilient people showing accelerated recovery from stressful encounters (Tugade & Fredrickson, 2004). Within his affective styles framework, Davidson (2004) considers resilience as one of the key dimensions and found that resilient persons exhibit higher baseline levels of left prefrontal activity (Jackson et al., 2003).

Extraversion and neuroticism. Extraversion and neuroticism are the two Big Five dimensions that have been shown to have a strong affective component and to be related to the frequency and intensity of positive and negative emotions, respectively (Schimmack, 2003). Recently, it was found that a similar relation holds for emotion duration with extraverts and neurotics experiencing long positive and negative emotions, respectively (Verduyn & Brans, 2012). A recent functional magnetic resonance imaging (fMRI) study further revealed that neurotics show prolonged amygdala activation following exposure to a negative event (Schuyler et al., 2012).

Depression. Depression is not only diagnostically characterized by enduring negative moods but also hampers emotional

recovery when faced with a negative event. In particular, Siegle and colleagues found that people with depression show sustained processing of negative information (Siegle, Granholm, Ingram, & Matt, 2001), and delayed amygdala recovery after being exposed to idiosyncratic negative stimuli (Siegle et al., 2002). Moreover, positive emotions are shorter in depressed populations with depressed individuals failing to sustain activation in neural circuits underlying positive affect and reward such as the nucleus accumbens (Heller et al., 2009).

Emotion regulatory actions. Emotion regulation refers to all the conscious and nonconscious strategies we use to increase, maintain, or decrease one or more components of an emotional response (Gross, 2007). As such, research on emotion regulation is by definition highly relevant for the study of emotion duration. Moreover, interindividual differences in emotion regulation may partially account for the predictive effect on emotion duration of the dispositions described before. For example, whereas neuroticism and depression have been associated with an inability to efficiently down-regulate negative emotions (Nolen-Hoeksema, 1991), resilient people flexibly adjust the level of emotional resources needed to meet the demands of the situation (Waugh, Wager, Fredrickson, Noll, & Taylor, 2008). In what follows, we describe the regulation strategies that have most consistently been found to be related to emotion duration.

Rumination. Rumination is defined as passively and repetitively focusing on one's symptoms of distress and their consequences (Nolen-Hoeksema & Morrow, 1993). This regulation strategy has been related to prolonged episodes of depression (Nolen-Hoeksema, 1991), as well as of anger and sadness (Verduyn et al., 2009). Rumination is typically studied in the context of negative emotions, but positive rumination following a positive event has been found to prolong the duration of positive emotions as well (Verduyn et al., 2011).

Reappraisal. Two types of reappraisal have been distinguished (Ochsner & Gross, 2008). The first variant of reappraisal refers to reinterpreting the emotion-provoking event. The second variant refers to distancing oneself from the event by adopting a detached, third-person perspective. Both types have most often been studied in the context of negative emotions and been shown to shorten their duration but in the case of positive events, reappraisal may shorten positive emotions as well (Verduyn, van Mechelen, Kross, et al., 2012; Verduyn et al., 2011).

Distraction. Distraction is defined as diverting one's attention away from the emotion-eliciting event (Gross, 2007). Distraction is the regulation strategy that is most often used in daily life (Brans, Koval, Verduyn, Lim, & Kuppens, 2013) and is an effective strategy to (temporarily) shut down an emotional episode (Freund & Keil, 2013), especially when the valence of the distractor is opposite to the valence of the emotion (Verduyn et al., 2011).

The neural mechanisms through which cognitive regulation strategies modulate the emotional response have attracted a lot of

research attention and space limitation prevents a complete review of this literature (for a recent review, see Ochsner & Gross, 2014). However, in these studies, the magnitude rather than the duration of neural activation is typically studied. Nevertheless, it is notable that whereas rumination was found to sustain amygdala activation over time (Siegle et al., 2002), reappraisal and distraction reduce amygdala activity while engaging prefrontal regions important for working memory, selective attention, and cognitive control more generally (McRae et al., 2010). Moreover, whereas reappraisal preferentially activates a network associated with processing affective meaning (including medial prefrontal cortex), distraction preferentially activates regions associated with the allocation of attention (including lateral prefrontal cortex; McRae et al., 2010).

As such, rumination may prolong emotions by strengthening the original affective meaning of the emotional event. In contrast, reappraisal may shorten emotions by attending to the emotion-provoking stimulus and changing its affective meaning, and distraction may shut down an emotion through decreased processing of the emotion-provoking event.

Conclusion

Emotions are dynamic processes of varying duration. Even though part of this variability may be due to dissimilarities in built-in time courses of different emotions or emotion components, emotion duration is mainly influenced by the events we encounter and the way we deal with them. Daily hassles do not perturb our emotional equilibrium for long but following highly impactful events, people elaborate on the meaning and consequences of the event. During this process, some people get stuck in vicious cycles of rumination causing the emotion to endure whereas others find the silver lining behind each dark cloud or manage to divert their attention towards something positive. Similarly, following positive events, some people initiate regulatory strategies to maintain their positive outlook whereas others fail to do so.

Future theoretical and empirical work is needed to advance our understanding of the duration of emotions. Theoretical work is necessary to sharply define the construct of emotion duration. The main complexity pertains to the end of an emotional response (for a more elaborate discussion on this, see van Mechelen et al., 2013) which has not been defined in a consistent manner across studies on emotion duration. Empirical research is needed to pinpoint at which stage and in which way the duration determinants modulate the unfolding of the emotional response. For this purpose the full intensity profile of the emotional response needs to be examined. This approach has only been adopted a few times (see e.g., Goldin et al., 2005, on the neural correlates of intensity profiles; and Heylen, Verduyn, van Mechelen, & Ceulemans, 2015, on the relation between appraisals, emotion regulation, and intensity profiles) but the results seem promising and call for follow-up studies.

Declaration of Conflicting Interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Note

- 1 Moods are often also differentiated from emotions based on a duration criterion with emotions being considered as shorter than moods (Beedie, Terry, & Lane, 2005). This relative order does, however, not imply that emotions are in absolute terms always or even typically short (Frijda et al., 1991).

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